

First Terminal Examination – 2082

Class : XI

Subject: Physics (1011)

Full Marks: 75

Time: 3 hrs.

SET 'A'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group A

Rewrite the best alternative of each question in your answer sheet. [11 × 1 = 11]

- The velocity of particle is given by $v = at + bt^2$ where 't' is time. The dimensions of b is
 a) $[LT^{-3}]$ b) $[LT^{-2}]$ c) $[LT^{-4}]$ d) $[LT^{-1}]$
- The number of significant figures in the measurement 0.010 cm is
 a) 1 b) 2 c) 3 d) 4
- $\vec{A}$ and $\vec{B}$ are two vectors such that $\vec{A} - \vec{B} = \vec{C}$ and $A + B = C$. Then the correct statement is
 a) $\vec{A}$ and $\vec{B}$ are in same direction
 b) $\vec{A}$ and $\vec{B}$ are in opposite direction
 c) $\vec{A}$ and $\vec{B}$ are orthogonal
 d) $\vec{A}$ and $\vec{B}$ are inclined at 45° to each other
- Identify the odd one:
 a) Temperature b) speed c) density d) displacement
- The angle between two forces of equal magnitude 'F' is 120° , then the magnitude of their resultant is
 a) $\frac{F}{\sqrt{2}}$ b) $\sqrt{2}F$ c) 2F d) F
- If the value of coefficient of superficial expansion of solid is 'n' times to that of its cubical expansion then the value of 'n' is
 a) 1/2 b) 1/3 c) 2/3 d) 3/2
- The mirror that used in vehicle for the rear view is:
 a) concave mirror b) convex mirror
 c) plane mirror d) parabolic mirror

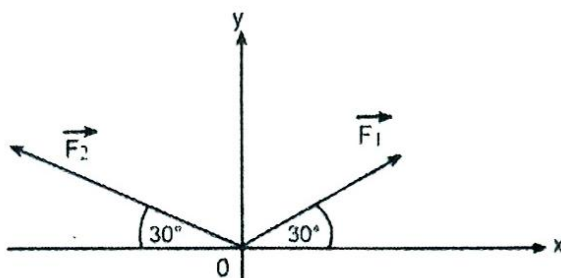
8. The magnification produced by a concave mirror in terms of focal length and object distance is:
 a) $\frac{u-f}{f}$ b) $\frac{u+f}{f}$ c) $\frac{f}{u-f}$ d) $\frac{f}{u+f}$
9. What is the time taken by a ray of light to emerge from a glass slab of thickness 2cm and refractive index 1.5?
 a) 10^{-8} s b) 10^{-9} s c) 10^{-10} s d) 10^{-12} s
10. If the refractive index of water is $4/3$ and that of given slab of glass immersed in it is $5/3$. What is the critical angle for a ray of light tending to go from glass to water?
 a) $\sin^{-1}(3/4)$ b) $\sin^{-1}(3/5)$ c) $\sin^{-1}(4/5)$ d) none
11. A child looks a floated fish in an aquarium just above from the water level of refractive index 1.33 and feel at 30cm below from the water. The fish is exactly at the depth of
 a) 30cm b) 35cm c) 40cm d) 45cm

Group – B

Give short answer to the following questions.

[8 × 5 = 40]

12. a) Write the dimensional formula for universal gravitational constant. [1]
 b) A student writes an expression of force causing a body of mass 'm' to move in a circular motion with a velocity 'v' as $F = mv^2$. Check its correctness using dimensional analysis. [2]
 c) The density of water is 1000 kg/m^3 at 4°C . Convert this value into g/cm^3 using dimensional analysis. [2]
13. a) Two forces $F_1 = 20 \text{ N}$ and $F_2 = 30 \text{ N}$, have direction as shown in the figure.



- i. Find the x and y-components of forces F_1 and F_2 . [2]
 ii. Find the magnitude and direction of the resultant of F_1 and F_2 . [2]

- b) If the magnitude of a vector is zero, is it meaningful to call it a vector? Explain. [1]
14. a) The energy of a photon is given by $E = hf$. Find the dimension and unit of Planck's constant h , where f is the frequency of radiation. [2]
- b) Differentiate between accuracy and precision during measurement. [1]
- c) A force (in Newton) expressed in vector notation as $\vec{F} = 4\hat{i} + 7\hat{j} - 3\hat{k}$ is applied on a body and produces a displacement (in meter) $\vec{S} = 3\hat{i} - 2\hat{j} - 5\hat{k}$ in 4 seconds. Estimate the power. [2]
15. a) Define Thermal equilibrium [1]
- b) Student claimed that thermometers are useless because a thermometer always registers its own temperature. How would you respond? [2]
- c) A faulty Fahrenheit thermometer shows a temperature of 206°F when immersed in boiling water. When a correct Celsius thermometer is dipped in the same water it records 96°C . Find the correction to be applied to the faulty thermometer when it records 206°F ? [2]
16. A brass rod of initial length ' l_1 ' undergoes linear expansion when it is heated from temperature θ_1 to θ_2 .
- a) Show that $l_2 = l_1(1 + \alpha\Delta\theta)$ where α is linear expansivity. [2]
- b) Does the value of α depends on initial length? Explain. [1]
- c) The volume of a metal sphere is increased by 1% of its original volume, when it is heated from 20°C to 222°C . Calculate the coefficient of superficial and cubical expansion of the metal. [2]
17. a) Define lateral shift and derive the expression of lateral shift produced by a glass-slab of thickness ' t '. [3]
- b) How much lateral shift is produced by a glass slab of thickness 5cm, when a light ray is incident on the surface with angle of incident 40° ? ($\mu_g = 1.33$) [2]
18. a) Establish the relation between relative refractive index and absolute refractive index. [3]
- b) A trunk of 4m long is laid along the principal axis of a convex mirror of focal length 1m. The end of the trunk nearer the mirror is 2 m from it. Find the length of the image of the trunk. [2]
19. a) Define critical angle and total internal reflection. [2]
- b) Write any two conditions for total internal reflection. [1]
- c) A light pipe ($\mu = 1.72$) is surrounded by a glass coating ($\mu = 1.5$). find the critical angle for the total internal reflection at the light pipe. [2]

Group – C

Give long answer to the following questions.

[3 × 8 = 24]

20. a) State the triangle law of vector addition. [1]
b) Using the triangle law of vector addition, find the magnitude and direction of the resultant of two vectors $\vec{P}$ and $\vec{Q}$ with an angle ' θ ' between them. [3]
c) A rocket fires two engines simultaneously. One produces a thrust of 725N directly forward, while the other gives a 513N thrust at 32.4° above the forward direction. Find the magnitude and direction of the resultant force that these engines exert on the rocket. [3]
d) Is a physical quantity having both magnitude and direction necessary a vector quantity? Explain. [1]
21. a) A circular metal plate has a small hole at its center is heated. What happens to the size of hole? Explain. [1]
b) For isotropic solid, show that its linear expansivity is approximately equal to half of superficial expansivity. [2]
c) Two bodies made of the same material have the same external dimensions and appearance, but one is solid and the other is hollow. Will their cubical expansivity be same? Explain. [2]
d) An aluminum rod when measured with a steel scale, both being at 25°C appears to be 1 m long. If the scale is correct at 0°C , what will be the length of the rod at 0°C ? (linear expansivity of aluminum = $26 \times 10^{-6} \text{ K}^{-1}$ and of steel = $12 \times 10^{-6} \text{ K}^{-1}$). [3]
22. a) Which mirror is shown in the figure? [1]
b) Define this mirror. [1]
c) Write any two uses of this mirror in our daily life. [1]
d) Which property does the mirror show, either converging or diverging? Illustrate it with figure. [2]
e) Derive the mirror formula using this mirror. [3]



❧ - The End - ❧